Signorini's Problem

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Abstract

This is the abstract.[1]

1 Introduction

This is the Introduction.

2 Mathematical Framework

2.1 Sobolev Spaces and the Trace Operator

In this subsection, we will make a brief excursion into selected topics in functional analysis, providing the necessary theory for the subsequent subsections. In classical analysis, derivatives are defined pointwise and require a certain level of smoothness. However, many functions of interest fall outside this framework. To extend the concept of differentiation to a broader class of functions, we can weaken what it means to *have a derivative*. The key idea is to characterize derivatives not by their pointwise behavior, but by how they act under integration against *test functions*. Using integration by parts, one can express the derivative of a smooth function in a form that remains meaningful even when the function itself is not differentiable in the classical sense. This leads naturally to the concept of weak derivatives, which allow us to treat integrable functions as *differentiable* in a generalized sense. We start by defining two important function spaces.

Definition 2.1 (Locally Integrable Functions)**.** *The space $L^1_{\mathrm{loc}}(\Omega)$ of all* `locally integrable` *functions on Ω is defined as*

$$L^1_{\mathrm{loc}}(\Omega) := \left\{ f : \Omega \to \mathbb{R} \text{ measurable} \;\middle|\; f|_K \in L^1(K)\ \forall K \subset \Omega,\ K \text{ compact} \right\}.$$

Definition 2.2. *The space $\mathcal{C}_c^\infty(\Omega)$ contains all infinitely often differentiable functions with compact support in Ω, i.e.*

$$\mathcal{C}_c^\infty(\Omega) = \{ f \in \mathcal{C}^\infty(\Omega) \mid \operatorname{supp}(f) \subset \Omega \text{ compact} \}$$

and $u \in \mathcal{C}_c^\infty(\Omega)$ is called a `test function`*.*

With these functions in hand, we can now extend the classical concept of differentiability to the concept of *weak* differentiability.

Definition 2.3 (Weak Derivatives)**.** *Let $u \in L^1_{\mathrm{loc}}(\Omega)$ and $\alpha \in \mathbb{N}_0^d$. A function $g \in L^1_{\mathrm{loc}}(\Omega)$ is called* `weak derivative` *of u of order α if*

$$\int_\Omega u\, \partial^\alpha \varphi \,\mathrm{d}x = (-1)^{|\alpha|} \int_\Omega g\, \varphi \,\mathrm{d}x \quad \forall \varphi \in \mathcal{C}_c^\infty(\Omega)$$

If such g exists, it is unique up to null sets and we write $\partial^\alpha u = g$ and keep the notation the same as in the strong sense of derivatives.

Of course, there are also spaces of weakly differentiable functions, named after the Soviet mathematician Sergei Lvovich Sobolev (1908–1989).

Definition 2.4 (Sobolev Spaces)**.** *For $k \in \mathbb{N}_0$ and $1 \le p \le \infty$ we define the* `Sobolev space` *$W^{k,p}(\Omega)$ to be the space of every L^p-function whose weak derivatives up to order k are also L^p-functions, i.e.*

$$W^{k,p}(\Omega) := \{ u \in L^p(\Omega) \mid \partial^\alpha u \in L^p(\Omega)\ \forall |\alpha| \le k \}$$

and we equip it with the norm

$$\|u\|_{W^{k,p}(\Omega)} := \left(\sum_{|\alpha| \le k} \|\partial^\alpha u\|^p_{L^p(\Omega)} \right)^{1/p}, \quad 1 \le p < \infty$$

and

$$\|u\|_{W^{k,\infty}(\Omega)} := \max_{|\alpha|\leq k} \|\partial^\alpha u\|_{L^\infty(\Omega)}.$$

The special case $p = 2$ provides us with a Hilbert space $H^k(\Omega) := W^{k,2}(\Omega)$ together with the inner product

$$(u, v)_{H^k(\Omega)} := \sum_{|\alpha|\leq k} \langle \partial^\alpha u,\, \partial^\alpha v\rangle_{L^2(\Omega)}$$

For the following purpose, we will have to consider the the space

$$H^1(\Omega) = \left\{ u \in L^2(\Omega) \;\middle|\; \nabla u \in L^2(\Omega; \mathbb{R}^d) \right\}$$

together with the induced norm

$$\|u\|_{H^1(\Omega)} = \left(\|u\|^2_{L^2(\Omega)} + \|\nabla u\|_{L^2(\Omega; R^d)} \right)^{1/2}$$

where the second L^2–term is understood as

$$\|\nabla u\|_{L^2(\Omega;\mathbb{R}^d)} = \left(\int_\Omega |\nabla u(x)|^2 \, \mathrm{d}x \right)^{1/2}.$$

Theorem 2.1 (Trace Theorem)**.**

3 Signorini's Problem

3.1 Physical and Geometric Setup of the Problem

We consider an elastic body occupying a reference configuration $\Omega \subset \mathbb{R}^d$, $d \in \{2,3\}$. Under the action of external forces and possible contact with a rigid foundation, the body deforms. In the *small–strain* framework, the deformation is described by the *displacement field* $u : \Omega \to \mathbb{R}^d$, and a material point $x \in \Omega$ is mapped to its deformed position

$$\varphi(x) := x' := x + u(x), \quad x \in \overline{\Omega}.$$

The aim of Signorini's problem is to determine an *equilibrium* displacement u under standard boundary conditions and *unilateral* contact constraints on the part of the boundary that may touch the rigid foundation. Throughout the whole report, $\Omega \subset \mathbb{R}^d$, $d \in \{2,3\}$ denotes an open, bounded and connected *Lipschitz* domain. The later means, that the boundary $\Gamma := \partial\Omega$ can locally be described as the graph of a Lipschitz continuous function. This ensures for example, that the outward normal vector $\mathrm{n}(x)$ is well–defined almost everywhere. The boundary is decomposed into three pairwise disjoint parts

$$\Gamma = \overline{\Gamma_D} \,\dot{\cup}\, \overline{\Gamma_N} \,\dot{\cup}\, \overline{\Gamma_C}$$

which are assumed to be open in Γ and where each part is subject to different boundary conditions to model different physical interactions (See Figure 1). On Γ_D, we apply *homogeneous Dirichlet* boundary conditions. This forces the body to be *clamped* and this part, i.e. the displacement should not act on Γ_D,

$$\forall x \in \Gamma_D : \; x + u(x) = x \quad \Longrightarrow \quad \mathrm{Tr}(u)|_{\Gamma_D} = 0.$$

The part Γ_N may be subject to a *Neumann* boundary condition, which means we impose a known *surface force density*. Γ_C denotes the *potential contact* boundary. We assume that Γ_C and Γ_D have positive $(d-1)$–Lebesgue measure. The rigid foundation is positioned so that any actual contact, both initially and in the deformed equilibrium configuration, may only occur on Γ_C. While Γ_C is prescribed a priori, the *active contact zone* $\Gamma_{\mathrm{act}}(u) \subset \Gamma_C$ is unknown and depends on the solution u. Later on, the boundary conditions imposed on Γ_C enforce *unilateral* contact through *nonpenetration* and *complementarity*.

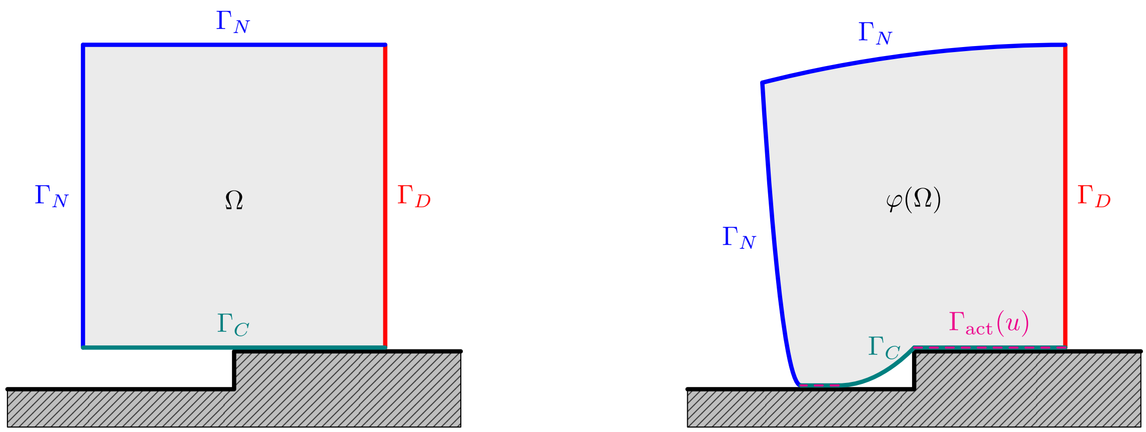

Figure 1: 2D example of Signorini's problem before (left) and after displacement (right).

3.2 Weak Formulation

3.2.1 Admissible Displacements

We start by introducing the energy space and the admissible set of displacements. Homogeneous Dirichlet boundary conditions on Γ_D are incorporated by

$$V := \left\{ v \in H^1(\Omega; \mathbb{R}^d) \;\middle|\; \mathrm{Tr}(v) = 0 \text{ on } \Gamma_D \right\}. \tag{3.1}$$

Let $\gamma = \mathrm{Tr} : H^1(\Omega; \mathbb{R}^d) \to H^{1/2}(\Gamma; \mathbb{R}^d)$ denote the trace operator. For $v \in V$ we define the normal component of the boundary displacement on Γ_C by

$$v_{\mathrm{n}} := (\gamma v) \cdot \mathrm{n} \quad \text{on } \Gamma_C,$$

where n is the outward unit normal on Γ. The unilateral nature of contact is expressed by the *nonpenetration* constraint. The rigid foundation is located on the exterior side of Γ_C in direction of n. Hence the body is not allowed to move across Γ_C into the obstacle. In the small–strain setting this yields

$$u_{\mathrm{n}} \leq 0 \quad \text{a.e. on } \Gamma_C.$$

This motivates the nonempty, closed and convex cone ([1, Prop. 3.1]) of admissible displacements

$$K := \{ v \in V \mid v_{\mathrm{n}} \leq 0 \text{ a.e. on } \Gamma_C \} \subset V. \tag{3.2}$$

3.2.2 Small Strain Elasticity

We start by defining the *small strain tensor* for a displacement function $u \in H^1(\Omega; \mathbb{R}^d)$, which is similar to the symmetric part of square matrices but for the gradient ∇u of u and through which the deformation of the body $\overline{\Omega}$ by u is quantified.

Definition 3.1 (Small Strain Tensor)**.** *The* `small strain tensor` ε *is defined by*

$$\varepsilon(u) := \frac{1}{2} \left(\nabla u + \nabla u^\top \right) \in L^2(\Omega; \mathrm{Sym}^2(\mathbb{R}^d)), \quad u \in H^1(\Omega; \mathbb{R}^d).$$

We denote by $\mathrm{Sym}^2(\mathbb{R}^d)$ the space of symmetric second order tensors,

$$\mathrm{Sym}^2(\mathbb{R}^d) := \left\{ s \in \mathbb{R}^d \otimes \mathbb{R}^d \;\middle|\; s^\top = s \right\} \cong \left\{ S \in \mathbb{R}^{d \times d} \;\middle|\; S^\top = S \right\}.$$

For $s, t \in \mathrm{Sym}^2(\mathbb{R}^d)$ we write denote by

$$s : t := \sum_{1 \leq i,j \leq d} s_{ij} t_{ij}$$

If $A(x)$ is a fourth order tensor and $s \in \mathrm{Sym}^2(\mathbb{R}^d)$, we use the *double contraction*

$$(A(x) : s)_{ij} := \sum_{1 \leq k,l \leq d} A_{ijkl}(x) s_{kl}, \quad s : A(x) : s := \sum_{1 \leq i,j,k,l \leq d} A_{ijkl}(x) s_{ij} s_{kl}.$$

In linear elasticity, stresses are assumed to depend linearly on strains. The material response is encoded in the fourth order *elasticity tensor* A, which maps a symmetric strain tensor to the corresponding *Cauchy stress tensor*. The tensor field A is prescribed by the material and may vary in space, describing a *heterogeneous* body.

Definition 3.2 (Elasticity tensor field). *An* `elasticity tensor field` *is a mapping*

$$A : \Omega \to \mathcal{L}(\mathrm{Sym}^2(\mathbb{R}^d); \mathrm{Sym}^2(\mathbb{R}^d))$$

such that the coefficients satisfy the symmetries

$$A_{ijkl}(x) = A_{jikl}(x) = A_{ijlk}(x) = A_{klij}(x), \quad 1 \le i,j,k,l \le d,$$

and the uniform ellipticity condition, i.e. there exists $\alpha_A > 0$ such that for a.e. $x \in \Omega$ and for all $s \in \mathrm{Sym}^2(\mathbb{R}^d)$,

$$s : A(x) : s \ge \alpha_A \, (s : s).$$

Moreover, there exists $C_A > 0$ such that

$$|A_{ijkl}(x)| \le C_A \qquad \text{for a.e. } x \in \Omega, \quad 1 \le i,j,k,l \le d.$$

The *internal* and *surface forces* in the body $\overline{\Omega}$ are described by the *Cauchy stress tensor*.

Definition 3.3 (Cauchy stress tensor). *Let A be an elasticity tensor field. For $u \in H^1(\Omega; \mathbb{R}^d)$ we define the* (linear) Cauchy stress tensor *as the field $\sigma(u) \in L^2(\Omega; \mathrm{Sym}^2(\mathbb{R}^d))$ given a.e. by*

$$\sigma(u)(x) := A(x) : \varepsilon(u)(x).$$

On V, we define the symmetric bilinear form

$$a(u, v) := \int_\Omega \sigma(u) : \varepsilon(v) \,\mathrm{d}x, \quad u, v \in V \tag{3.3}$$

representing the *virtual work of internal forces* and the linear bounded functional

$$\ell(v) := \int_\Omega f \cdot v \,\mathrm{d}x + \int_{\Gamma_N} F \cdot \gamma v \,\mathrm{d}s, \quad v \in V, \tag{3.4}$$

with *volume force density* $f \in L^2(\Omega; \mathbb{R}^d)$ on the body and *surface loads* $F \in L^2(\Gamma_N; \mathbb{R}^d)$ on the Neumann boundary, representing the *virtual work of external forces*. In the following, we want to prove standard properties of a and ℓ. For this, we will need the following inequality, which is a result of Korn's second inequality ([1, Theorem 3.3]) and the *Peetre–Tartar* Lemma ([1, Lemma 3.5]).

Theorem 3.1. *Assuming $|\Gamma_D| > 0$, there exists a constant $C_K > 0$ such that*

$$\|v\|_{H^1(\Omega)} \le C_K \|\varepsilon(v)\|_{L^2(\Omega)} \quad \forall v \in V.$$

Corollary 3.1. *The norms $\|\varepsilon(\cdot)\|_{L^2(\Omega)}$ and $\|\cdot\|_{H^1(\Omega)}$ are equivalent on V with*

$$\frac{1}{C_K} \|v\|_{H^1(\Omega)} \le \|\varepsilon(v)\|_{L^2(\Omega)} \le \|v\|_{H^1(\Omega)} \quad \forall v \in V.$$

Proof. The first inequality follows directly form Theorem 3.1. Furthermore, we have Furthermore, for all $v \in V$ we have

$$|\varepsilon(v)(x)| = \left| \frac{1}{2} (\nabla v(x) + \nabla v^\top) \right| \le \frac{1}{2} (|\nabla v(x)| + |\nabla v(x)^\top|) = |\nabla v(x)|$$

and thus

$$\|\varepsilon(v)\|^2_{L^2(\Omega)} = \int_\Omega |\varepsilon(v)(x)|^2 \,\mathrm{d}x \le \int_\Omega |\nabla v(x)|^2 \,\mathrm{d}x = \|\nabla v\|^2_{L^2(\Omega)} \le \|v\|^2_{H^1(\Omega)},$$

where the last inequality holds due to the definition

$$\|v\|^2_{H^1(\Omega)} = \|v\|^2_{L^2(\Omega)} + \|\nabla v\|^2_{L^2(\Omega)}.$$

■

Lemma 3.1. *The symmetric bilinear form a defined in (3.3) is*

i) bounded, i.e. there exists $C_1 < \infty$ such that

$$|a(u,v)| \leq C_1 \|u\|_{H^1(\Omega)} \|v\|_{H^1(\Omega)} \quad \forall v, w \in V,$$

ii) V-elliptic, i.e. there exist $C_2 > 0$ such that

$$a(v,v) \geq C_A \|v\|^2_{H^1(\Omega)} \quad \forall v \in V \setminus \{0\}.$$

Moreover, the linear functional ℓ defined in (3.4) is bounded, i.e. there exists $C_3 > 0$ such that

$$|\ell(v)| \leq C_3 \|v\|_{H^1(\Omega)}.$$

Proof. Let $v, w \in V$ be arbitrary. Then

$$|a(v,w)| \leq \|A\varepsilon(v)\|_{L^2(\Omega)} \|\varepsilon(w)\|_{L^2(\Omega)}$$

by the Cauchy–Schwarz inequality. With Definition 3.2 there exists $C_A > 0$ such that

$$\|A\varepsilon(v)\|_{L^2(\Omega)} \leq C_A \|\varepsilon(v)\|_{L^2(\Omega)}.$$

Combining this with the second inequality form Corollary 3.1 we have

$$|a(v,w)| \leq C_A \|\varepsilon(v)\|_{L^2(\Omega)} \|\varepsilon(w)\|_{L^2(\Omega)} \leq C_A \|v\|_{H^1(\Omega)} \|w\|_{H^1(\Omega)}.$$

The ellipticity of a follows by

$$a(v,v) = \int_\Omega \varepsilon(v) : A : \varepsilon(v) \, \mathrm{d}x \geq \alpha_A \int_\Omega \varepsilon(v) : \varepsilon(v) = \alpha_A \|\varepsilon(v)\|^2_{L^2(\Omega)},$$

where $\alpha_A > 0$ exists due to Definition 3.2. The first inequality of Corollary 3.1 finally yields

$$a(v,v) \geq \alpha_A \|\varepsilon(v)\|^2_{L^2(\Omega)} \geq \alpha_A \left(\frac{1}{C_K} \|v\|_{H^1(\Omega)} \right)^2 = \frac{\alpha_A}{C_K^2} \|v\|^2_{H^1(\Omega)}.$$

Again by Cauchy–Schwarz inequality,

$$|\ell(v)| \leq \|f\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} + \|F\|_{L^2(\Gamma_N)} \|\gamma v\|_{L^2(\Gamma_N)} \quad \forall v \in V.$$

By the Trace Theorem 2.1 there exists $C_{\mathrm{tr}} > 0$ such that

$$\|\gamma v\|_{L^2(\Gamma_N)} \leq \|\gamma v\|_{L^2(\Gamma)} \leq \|\gamma v\|_{H^{1/2}(\Gamma)} \leq C_{\mathrm{tr}} \|v\|_{H^1(\Omega)} \quad \forall v \in V.$$

Finally,

$$|\ell(v)| \leq (\|f\|_{L^2(\Omega)} + C_{\mathrm{tr}} \|F\|_{L^2(\Gamma_N)}) \|v\|_{H^1(\Omega)}.$$

■

3.2.3 Energy Minimization

The total potential energy of the elastic body is given by the linear bounded functional

$$\mathcal{J}(v) := \frac{1}{2} a(v,v) - \ell(v), \quad v \in V. \tag{3.5}$$

In the presence of unilateral contact, the admissible displacements are restricted to the closed convex set $K \subset V$. We therefore consider the following constrained minimization problem.

Problem 3.1 (Signorini's Problem as Energy Minimization). *Find $u \in K$ such that*

$$\mathcal{J}(u) = \min_{v \in K} \mathcal{J}(v).$$

We now what to prove well-posedness of the above Problem 3.1. This consist of existence and uniqueness of a solution and continuous dependency of the solution on the load functional ℓ. We start by *coercivity* and *weakly lower semicontinuity* of the functional $\mathcal{J}$ to be minimized over K

Proposition 3.1 (Weakly lower semicontinuity of $\mathcal{J}$ on V). *The functional $\mathcal{J}$ defined in (3.5) is weakly lower semicontinuous on V, i.e. if $v_n \rightharpoonup v$ in V, then*

$$\mathcal{J}(v) \leq \liminf_{n \to \infty} \mathcal{J}(v_n).$$

Proof. For $s \in L^2(\Omega; \mathrm{Sym}^2(\mathbb{R}^d))$ we define the norm

$$\|s\|_A^2 := \int_\Omega s : A : s \, \mathrm{d}x,$$

such that $a(v, v) = \|\varepsilon(v)\|_A^2$. Furthermore, $v \mapsto \varepsilon(v)$ defines a linear and bounded operator $V \to L^2(\Omega; \mathrm{Sym}^2(\mathbb{R}^d))$. Assuming that $v_n \rightharpoonup v$ in V, we have

$$\varepsilon(v_n) \rightharpoonup \varepsilon(v) \quad \text{in } L^2(\Omega; \mathrm{Sym}^2(\mathbb{R}^d))$$

and noticing that every norm is weakly lower semicontinuous, we have

$$a(v, v) = \|\varepsilon(v)\|_A^2 \leq \liminf_{n \to \infty} \|\varepsilon(v_n)\|_A^2 = \liminf_{n \to \infty} a(v_n, v_n).$$

Thus, we conclude

$$\mathcal{J}(v) = \frac{1}{2} a(v, v) - \ell(v) \leq \frac{1}{2} \liminf_{n \to \infty} a(v_n, v_n) - \lim_{n \to \infty} \ell(v_n) = \liminf_{n \to \infty} \mathcal{J}(v_n),$$

where $\ell(v_n) \to \ell(v)$ because $\ell \in V^*$. ■

Proposition 3.2 (Strict Convexity of $\mathcal{J}$). *The Functional $\mathcal{J}$ defined in (3.5) is strictly convex on V.*

Proof. For $u, v \in V$ and $t \in [0, 1]$, we set $w_t := (1 - t)u + tv$. Then

$$a(w_t, w_t) = (1 - t)a(u, u) + ta(v, v) - t(1 - t)a(u - v, u - v)$$

and

$$\ell(w_t) = (1 - t)\ell(u) + t\ell(v).$$

Together, we have

$$\mathcal{J}(w_t) = (1 - t)\mathcal{J}(u) + t\mathcal{J}(v) - \frac{1}{2} t(1 - t)a(u - v, u - v).$$

If $u \neq v$ then $u - v \neq 0$ and thus by V-ellipticity of a we have $a(u - v, u - v) > 0$ and $t(1 - t) > 0$. This implies

$$\mathcal{J}(w_t) < (1 - t)\mathcal{J}(u) + t\mathcal{J}(v).$$

■

Theorem 3.2 (Well-posedness of Problem 3.1)**.** *There exists a unique minimizer $u \in K$ of $\mathcal{J}$ and it holds*

$$\mathcal{J}(v) - \mathcal{J}(u) \geq \frac{\alpha}{2}\|v-u\|_V^2 \quad \forall v \in K,$$

Consequently, Problem 3.1 has a unique solution and error norms are controlled by energy differences.

3.2.4 Variational inequality as an optimality condition

3.2.5 Existence and Uniqueness

3.3 Strong Formulation

3.4 Mixed Formulation

3.4.1 Mixed Weak Foram as Sattlepoint

3.4.2 Well–posedness and inf–sup condition

4 Outlook

A References

[1] Franz Chouly, Patrick Hild, and Yves Renard. *Finite Element Approximation of Contact and Friction in Elasticity*, volume 48 of *Advances in Continuum Mechanics*. Springer International Publishing, 2023.

B List of Figures

1 2D example of Signorini's problem before (left) and after displacement (right). . 5